

$$= 1.9 \text{ m/s}$$

11. Ans: B

$$\begin{aligned}\text{Total Energy } E_T &= \frac{1}{2} m \omega^2 x_o^2 \\ &= \frac{1}{2} m (2\pi/T)^2 x_o^2 \\ &= \frac{1}{2} (0.60)(2\pi/5.0)^2 (0.15)^2 \\ &= 0.0107 = 1.1 \times 10^{-2} \text{ J}\end{aligned}$$